

DSP Assignment 1

Chinmay Kulkarni 23110179, Rupanjana Ghosh 23110282, Sachin Kumar Jha 23110286

October 2025

1 Objective

Secure and Intelligent Speech Communication with Motion-Sensor Integration.

2 Audio Encryption-Decryption and Transfer Process

2.1 Main Idea

- Audio has been taken by the microphone module in Simulink.
- Since the audio is stereo, hence each instance has 2 channels, right and left.
- The fft of the audio contains complex numbers,
- The idea of scrambling is to get a circular shifted version of the fft bins, which is controlled by the parameter k in the function.
- The scrambled output is sent through UDP protocol channels and fetched by the other phone.
- The distorted audio will be unscrambled in the other phone.
- After unscrambling, an inverse Fourier transform is performed and the original signal is returned.
- The gyroscope and accelerometer data is sent through separate UDP channels and is output at the other phone and plotted.

2.2 Block Explanation

2.2.1 Phone A

- Microphone Module

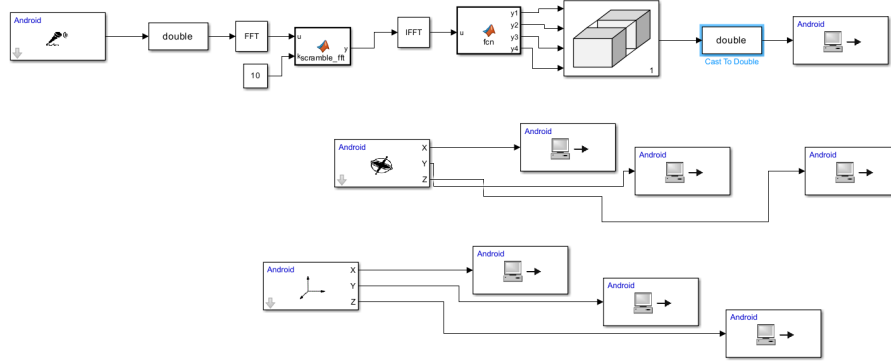


Figure 1: Complete Simulink Model of Phone A



Figure 2: Microphone Phone A

This module captures the audio from the microphone of phone A and stores it in a 256X2 format, 256 being the number of samples and 2 channels since the audio is stereo. the datatype is converted to double for later convenience while transmission. The Sampling frequency of the microphone is chosen to be **8 kHz**. Even though the usual audio sampling rate is 44.1 kHz, it has been significantly reduced as the **processor handling capacity could only deliver for such low sampling rates**. Also, **this does not cause any significant issue in sampling, as our voice box is not capable of generating frequencies above 8 kHz**.

- FFT and Scramble Module

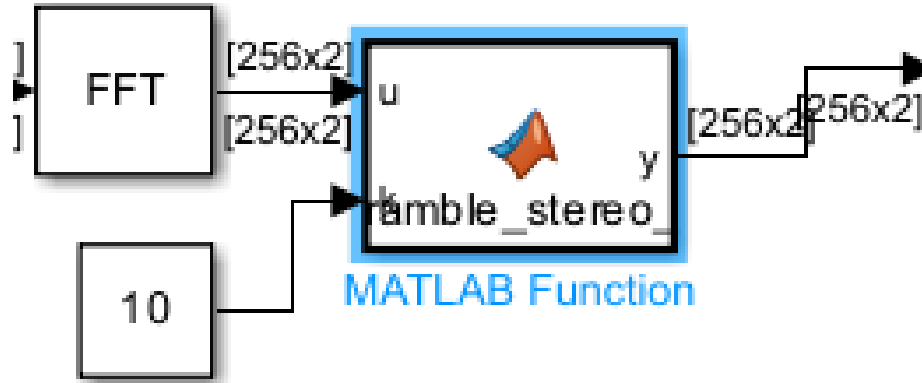


Figure 3: FFT and Scramble Module

The input from the microphone is passed on to the Simulink in-built fft module which extracts frequency components from the data. This data is complex and an 256X2 matrix containing of complex values is got as an output. This FFT data is passed through the scramble module which is user-defined to scramble the frequency components.

- Key Based Scrambling Operation

```

function y = scramble_stereo_fft(u, k)
    left_channel_fft = u(:, 1);
    right_channel_fft = u(:, 2);

    scrambled_left = scramble_channel(left_channel_fft, k);
    scrambled_right = scramble_channel(right_channel_fft, k);

    y = [scrambled_left, scrambled_right];
end

function scrambled_ch = scramble_channel(ch, k)
    N = length(ch);
    scrambled_ch = complex(zeros(N, 1));
    shift_amount = k*13;

    for i = 1:N
        new_i = mod(i - 1 + shift_amount, N) + 1;
        scrambled_ch(new_i) = ch(i);
    end
end

```

Figure 4: Scrambling Operation

The key here used to scramble lies in the variable k , and the shift amount is chosen to be $k*13$. Thus, for a k , the whole frequency matrix will be circular shifted by the shift amount. Such is the scrambling operation. `scrambled_ch` is the new scrambled matrix which is the output of this function. The values are all complex. `scramble_stereo_fft` function arranges the left and right channels and `scramble_channel` function scrambles the data.

- IFFT and Transfer to Phone B

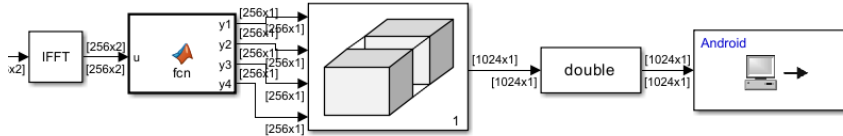


Figure 5: IFFT and Transfer

The scrambled frequency matrix is now passed to an IFFT block and a distorted sound is created, It also contains complex numbers. The real

and imaginary parts of both the channels have been segregated in **fcn** block. Since UDP transfer and only happen with one column matrices, they are again concatenated by the Matrix concatenation block and finally transferred to phone B via UDP protocol.

- Gyroscope and Accelerometer Data

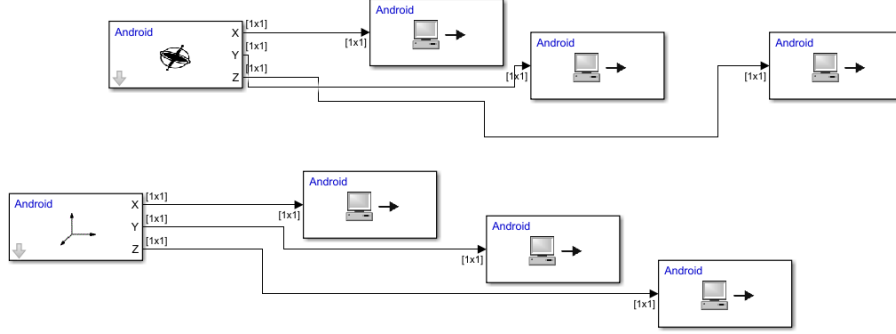


Figure 6: Gyroscope and Accelerometer Data

The in-built simulink blocks for Accelerometer and Gyroscope data collection have been taken, each have three axes which output a single data. All the data, six in total, have been passed separately through 6 UDP send blocks.

2.2.2 Phone B

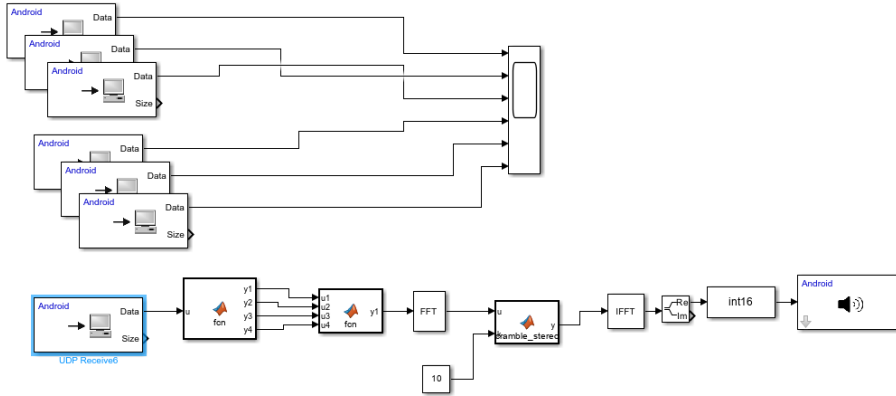


Figure 7: Complete Simulink Model of Phone B

- UDP Receive and Combination

```

function y1 = fcn(u1,u2,u3,u4)
    y1 = [u1+1j*u3, u2+1j*u4];

end

```

Figure 10: Making the matrix

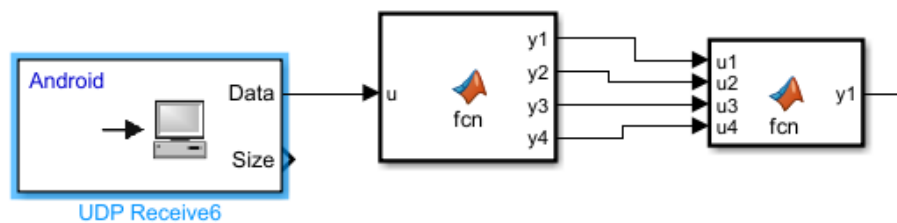


Figure 8: UDP Receive and Combination

```

function [y1, y2, y3, y4] = fcn(u)
    y1 = u(1:256);
    y2 = u(257:512);
    y3 = u(513:768);
    y4 = u(769:1024);

end

```

Figure 9: Separating the concatenated input

The scrambled audio data is received. All the four components, [1:256],

[257:512],[513:768],[769:1024]; which are the real and complex parts which were concatenated are separated out. Combining the respective real and complex parts, it is reconstructed into a 2 column matrix. The sample time of the UDP block is $\frac{FrameSize}{SamplingFrequency} = 0.032$.

- Descrambling

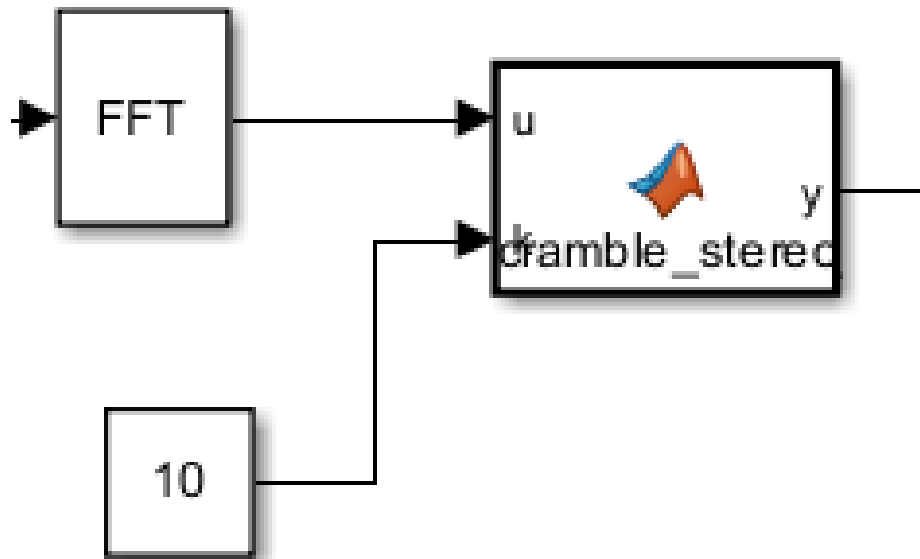


Figure 11: Descrambling

```

function y = descramble_stereo_fft(u, k)
    left_channel_scrambled = u(:, 1);
    right_channel_scrambled = u(:, 2);

    descrambled_left = descramble_channel(left_channel_scrambled, k);
    descrambled_right = descramble_channel(right_channel_scrambled, k);

    y = [descrambled_left, descrambled_right];
end

function descrambled_ch = descramble_channel(ch, k)
    N = length(ch);
    descrambled_ch = complex(zeros(N, 1));
    shift_amount = k*13;

    for i = 1:N
        new_i = mod(i - 1 - shift_amount, N) + 1;
        descrambled_ch(new_i) = ch(i);
    end
end

```

Figure 12: Descrambling Operation

The FFT from distorted signal has been passed through the fft block and descrambled as an inverse operation of the scrambling function.

- IFFT and Output Audio



Figure 13: Output Audio

The inverse Fourier transform of the fft has been calculated and fed to the output audio block. This gives us our clean input signal back.

- Gyroscope and Accelerometer Output

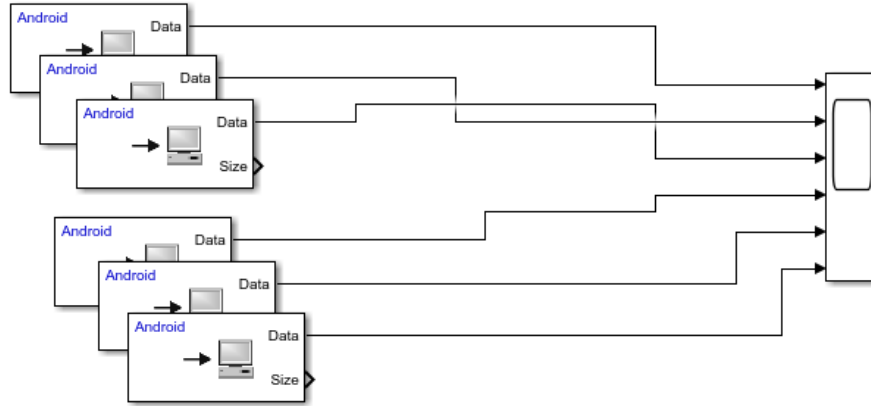


Figure 14: Gyroscope and Accelerometer Output

The gyroscope and accelerometer data is put through a scope and plotted on the output.

3 Challenges Faced

- Sampling frequency had to be decreased from 44 kHz as the bits were not transferred properly.
- There was interference when all the signals were transferred together through a single UDP channel by concatenating them. Hence, separate channels for audio, gyroscope and accelerometer were chosen.

4 Conclusion

- The input was unintelligible when the keys were not synchronised in the phones. Hence the scrambling worked only when the keys of both the phones matched.
- Even though the sound on the receiver was distinguishable, it had a minor noise component.
- There was some time delay between the audio input and output, which is due to the transmission protocol taking time to pass the bits.
- Overall, the objective of the exercise which was to encrypt and decrypt an audio signal has been attempted.